2025년 경상국립대학교 RISE 사업단 [RISE-AI]
생성형 AI+X 활용 아이디어 경진대회 제안서

Ⅰ 팀 구성

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※ 숙소 배정을 위한 성별 표기

Ⅱ AI+X의 정의

X의 정의(외국인 지원, 교육, 창업, 자동화, 생산성 등)와 선택한 이유

X = UAV (Unmanned Aerial Vehicle) Pre-flight Safety Analysis

UAV safety is a **pressing social and industrial problem**, not merely a technical issue. As UAVs are increasingly deployed in logistics, infrastructure inspection, disaster response, and urban environments, the frequency and impact of safety incidents are rising globally.

- **Increase in UAV-related incidents**: According to the FAA Drone Sightings Near Airports dataset, thousands of UAV-related safety sightings are reported annually, including abnormal flight behavior, airspace intrusions, and near-miss situations with manned aircraft.
- **High societal impact**: UAV accidents can cause human injuries, damage critical infrastructure, and significantly reduce public trust in autonomous aerial systems.
- **Regulatory pressure**: Aviation authorities such as the FAA and EASA increasingly require **risk-based safety assessments before flight approval**, especially for BVLOS (Beyond Visual Line of Sight) operations.

Despite this urgency, **pre-flight UAV safety analysis remains challenging** due to the following characteristics:

- Safety knowledge is mainly stored as **unstructured incident narratives**, which are difficult to reuse systematically.
- Hazard interactions are **context-dependent and non-linear**, making them hard to model with fixed rules.
- Traditional safety approaches rely on **static rule-based or parametric models**, which struggle to adapt to new environments or rare scenarios.

These characteristics make UAV safety a domain where **AI convergence is necessary rather than optional.**

In particular, **Generative AI (Large Language Models, LLMs)** is well-suited to this domain because it can:

- Reason over **real-world incident cases** expressed in natural language,
- Convert implicit safety knowledge into **structured hazard scenarios**, and
- Provide **human-understandable safety explanations**, which are essential in safety-critical systems.

Ⅲ 제안 내용 (1차 제출 후 대회 기간 전까지 수정 가능)

※ 글, 그림, 코드 등을 활용하여 작성, 장 수에 구애받지 않고 작성

	내용
제안 배경	◆ Civilian UAV operations are rapidly expanding in shared airspace. ◆ UAV safety sighting and near-miss data are publicly available (e.g., FAA Drone Sightings Near Airports), but remain underutilized for proactive **pre-flight safety planning**. ◆ Current UAV safety practices focus primarily on **post-incident analysis**, rather than **systematic pre-flight hazard anticipation**. Existing UAV safety analysis approaches exhibit several limitations: ◆ Simulation-based testing exists, but **scenario selection is manual and heavily dependent on expert intuition**. ◆ Incident databases are **not directly executable**, making it difficult to translate past accidents into testable scenarios. ◆ Safety engineers must analyze **large volumes of incident reports** without automated reasoning support. As a result, **rare but high-impact hazard combinations** are often overlooked, particularly when UAVs are deployed in new environments or mission profiles. Recent advances in LLMs enable: ◆ Context-aware reasoning over large-scale textual data, ◆ Generation of **structured outputs under strict constraints**, and ◆ Integration with external tools such as simulators and analytics engines. These capabilities make it possible to **bridge real-world safety knowledge and executable pre-flight testing**, which was difficult to achieve using previous AI or rule-based methods.
해결하려는 문제	This project focuses specifically on **pre-flight UAV safety analysis**. Current pre-flight safety processes suffer from: ◆ Manual and experience-driven hazard identification, ◆ Limited reuse of historical incident knowledge, ◆ Difficulty in converting safety knowledge into **systematic, testable scenarios**. ◆ Critical hazards may not be tested before deployment. ◆ Safety quality depends heavily on individual operator expertise. ◆ UAV missions face increased risk of unexpected failures. There is a strong need for a system that: ◆ Systematically transforms real-world incidents into executable safety scenarios,

	내용
	◆ Evaluates mission risk **before flight**, and ◆ Provides **interpretable safety insights** to human operators.
생성형AI 활용 아이디어	We propose **AeroGuardian**, an AI-based **pre-flight UAV safety analysis framework**. In AeroGuardian, LLMs are used **not as free-text generators**, but as **reasoning agents** with clearly defined roles: ◆ Interpreting real-world UAV safety sighting and near-miss narratives, ◆ Generating **schema-constrained hazard scenarios** suitable for simulation, ◆ Explaining safety risks in human-readable language. **Input** ◆ UAV mission configuration, ◆ Hazard scenario templates, ◆ Retrieved UAV incident cases. **Processing** ◆ Retrieval-Augmented Generation (RAG) to ground reasoning in real-world UAV safety sighting and near-miss data, ◆ Schema-guided scenario generation using DSPy*, ◆ Simulation-based stress testing, ◆ Data-driven risk analysis using DL and ML models. **Output** ◆ Mission-level risk classification (Low / Medium / High), ◆ Human-readable safety explanation reports for operators. (*DSPy: a framework for programmatic optimization of LLM prompts.) Compared to traditional safety analysis approaches: ◆ AeroGuardian **automatically links real incident knowledge to simulation-based testing**, ◆ LLM-based reasoning enables **context-aware and adaptive scenario generation**, ◆ The system provides **explainable, data-driven safety decisions**, not just risk scores.
구현 방안 / 해결 방안	AeroGuardian follows a **modular cognitive pipeline**, where each component has a single, verifiable responsibility: 1. Safety Knowledge Retrieval Module (RAG) 2. LLM-based Scenario Planning Module 3. UAV Simulation Module (PX4 SITL) 4. Anomaly Detection Module (Deep Learning) 5. Risk Classification Module (Machine Learning) 6. Safety Explanation Module (LLM) Each module operates with clearly defined inputs and outputs to ensure reproducibility and traceability. ◆ **Phase 1**: Incident data collection and preprocessing ◆ **Phase 2**: Scenario schema design and LLM prompting ◆ **Phase 3**: UAV simulation and telemetry data collection ◆ **Phase 4**: Risk modeling and validation ◆ **Phase 5**: Safety explanation and reporting

	내용
기대효과	This project is expected to deliver the following value: ◆ Improved **pre-flight UAV safety** through systematic and data-driven hazard testing, ◆ Reduced dependence on manual expert knowledge, ◆ Increased trust in AI-assisted safety systems through explainability, ◆ A reusable and extensible framework for **AI-driven safety analysis** in autonomous systems.

[서식]

개인정보 수집 · 이용 동의서

경상국립대학교 RISE사업단 RISE-AI 과제(이하 사업단)에서는 「생성형 AI+X 활용 아이디어 경진대회」 운영을 위하여 귀하의 개인정보를 아래와 같이 수집·이용하오니 동의 여부를 체크하여 주시기 바랍니다.

[수집 · 이용하는 개인정보의 항목]

사업단은 아래와 같은 개인정보를 수집 · 이용하고 있습니다.

- **[항목]** 성명, 소속, 학번, 전화번호, 이메일, 사진 등

[개인정보 수집 · 이용 목적]

사업단은 아래와 같은 목적을 위해 개인정보를 수집 · 이용합니다.

- 「생성형 AI+X 활용 아이디어 경진대회」 공지사항 등 안내 및 운영

[개인정보 수집 및 보유기간]

- 수집 및 보유근거: 「개인정보보호법 제15조에 의함」
- 보유기간 : 5년

[동의 거부 시]

개인정보 수집 동의에 거부하실 수 있으며 다만, 거부 시 **프로그램 참가 활동이 제한**될 수 있습니다.

[필수항목]에 대한 개인정보의 수집 및 이용에 동의하십니까?

☑ 동의함 ☐ 동의하지 않음

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